Experimentation & Model Training:
Predicting Term Deposit Subscriptions for the Bank Marketing Campaign

Course: GAI 601 — Machine Learning **Assignment 3:** Experimentation & Model Training

The assignment required experimentation with a Decision Tree (DT) algorithm on the bank marketing dataset (39,404 customers, only 11.7% subscribed — a severe class imbalance). I conducted six DT experiments as the core requirement, then extended the analysis with six experiments each on Random Forest (RF) and XGBoost to compare different ways of combining trees. For every experiment I followed the rubric's five-step loop: hypothesis, single change, training, evaluation, recommendation. Because of the imbalance, accuracy was misleading (predicting "no" for everyone scores 88.3%), so I tracked F1 score, which balances precision and recall on subscribers, with ROC-AUC as a complementary ranking measure. The no-optimization rule was important: changing one setting at a time let me isolate exactly *why* each change helped or hurt.

The unconstrained DT baseline (DT-1) overfit severely (training accuracy 99% vs test 83%, F1 = 0.329). Hypothesizing that this was variance, I limited tree depth to 5 (DT-2) — which closed the overfit but made the model too cautious, missing 80% of subscribers (F1 = 0.317). I then swapped depth for a softer leaf-size rule of 50 samples per leaf (DT-3), preserving more signal (F1 = 0.358). Switching the splitting criterion from gini to entropy (DT-4) had almost no effect (F1 = 0.367), a useful lesson that not every setting changes the outcome. The largest improvement came from balanced class weighting (DT-5), which lifted F1 to 0.413 by raising recall from 0.25 to 0.68. Removing a redundant age-group feature (DT-6) held F1 at 0.411. **Among the six DT experiments, DT-5 was the winner with F1 = 0.413, precision = 0.297, recall = 0.678.**

Extending to RF and XGBoost, I applied the same lessons. Class weighting alone in RF (RF-2) actually slightly hurt F1, because deep trees still memorized despite focusing on the rare class — a critical insight that imbalance handling and regularization must work together in ensembles. Pairing class weighting with depth control of 10 on 300 trees (RF-4) gave the strongest result. Among the six RF experiments, RF-4 was the winner with F1 = 0.486, precision = 0.400, recall = 0.620, and ROC-AUC = 0.800. XGBoost with the same imbalance handling, a lower learning rate of 0.1, and shallower trees of depth 4 (XGB-4) performed essentially as well, limited only by the no-optimization rule that prevented tuning multiple settings together. Among the six XGBoost experiments, XGB-4 was the winner with F1 = 0.477, precision = 0.379, recall = 0.644.

Across all three algorithms compared, RF-4 was the overall winner. RF won because its averaging across many trees produces robust results without aggressive tuning — exactly the strength rewarded by the no-optimization constraint. The deepest takeaway was methodological: disciplined experimentation — change one thing, measure, reflect — was itself the most valuable skill, more than any single algorithm or number.